

A Simple Timer Based On Arduino Uno

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Presented here is a simple timer built around an Arduino Uno board. It can be used to measure the time delay duration of devices like timer relays, circuit breakers, and other sequential logic circuits. This timer will be a low-cost solution for

laboratory/industrial requirements. The author's prototype is shown in Fig. 1.

Circuit and working

The schematic diagram of the kit, shown in Fig. 2, has few components besides the Arduino Uno board (Board1). It has external input contacts to start and stop the timing measurement. Both the contacts can be either normally open (N/O) or normally closed (N/C). The change-over operation of start contacts initiates timing measurement, and of the stop contacts stops the measurement.

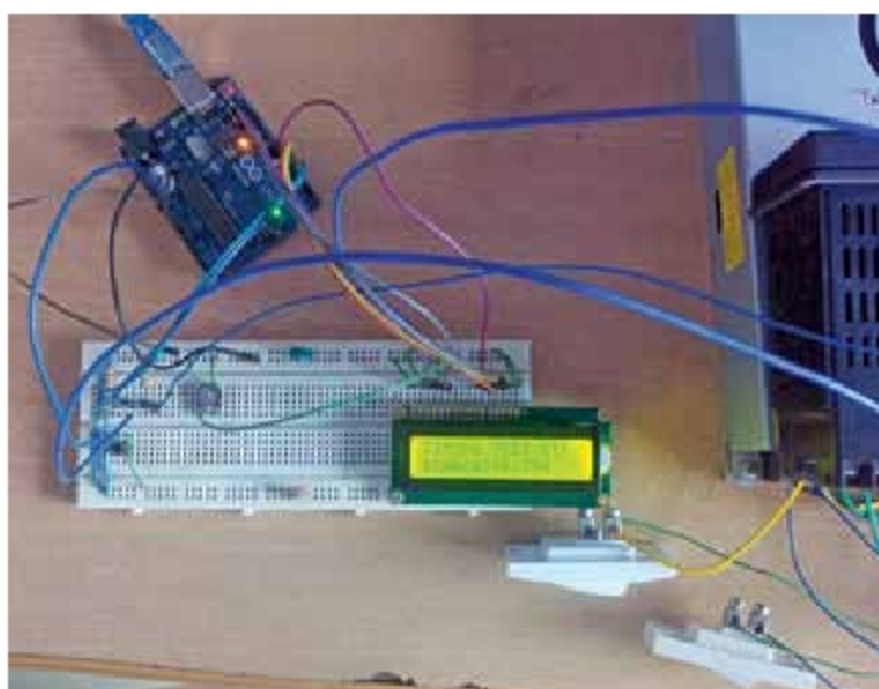


Fig. 1: Author's prototype of the timer kit

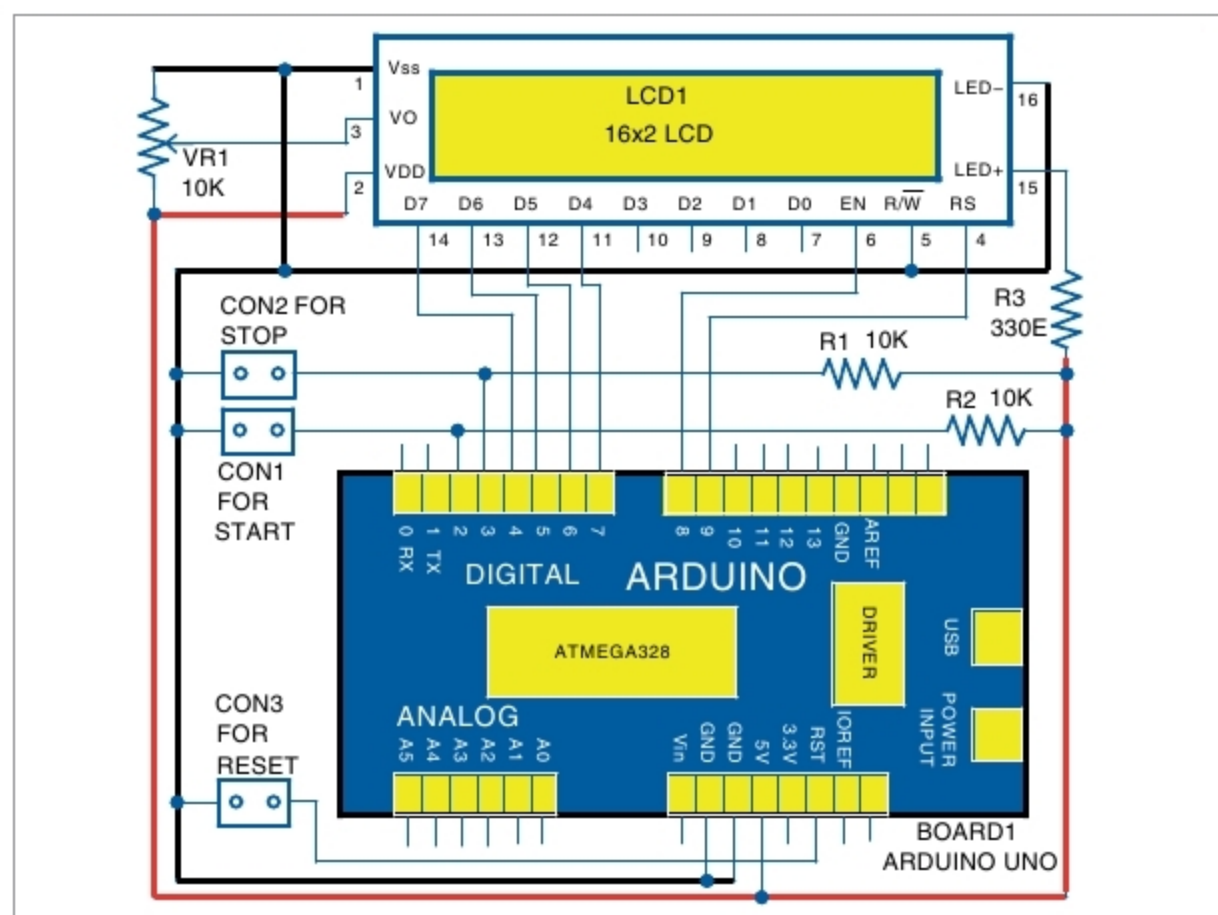


Fig. 2: Schematic diagram of the timer kit

The measured time is displayed in seconds on LCD1. Measurement in the range of milliseconds to several minutes is possible. It is tested with known values set in a mini programmable logic controller (PLC.) A reset button (connected across CON3) is provided to reset the display to zero and start a fresh timing measurement.

The software (timing_kit.ino) is written in Arduino programming language using Arduino IDE version 1.8.5. The sketch/program uses the interrupt and millis () functions of the Arduino. Ensure that you select the correct board from the Tools menu in Arduino IDE before uploading the timing_kit.ino sketch file to the board.

Construction and testing

Assemble the circuit on a breadboard or PCB. An actual-size PCB layout for the timer kit is shown in Fig. 3 and its components layout in Fig. 4.

PARTS LIST

Semiconductors:

Board1 - Arduino Uno

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 10-kilo-ohm

R3 - 330-ohm

VR1 - 10-kilo-ohm potmeter

Miscellaneous:

LCD1 - 16x2 LCD

CON1-CON3 - 2-pin connector

- Push-to-on reset switch

- On/off switch (2 numbers for testing)

- USB cable

EFY Note

The source code of this project is available for free download at source.efymag.com